Solution 1: Forward and Backward chaining are inference techniques used in rule based expert systems to derive solutions from a knowledge base.

1. Forward Chaining (Data-driven Reasoning)

Forward chaining starts with known facts and applies inference rules to generate new facts until a goal is reached.

Working:

1. Collect all known facts
2. Apply rules whose conditions match these facts.
3. Generate new facts.
4. Repeat until the goal is found or new fact can be derived.

Example:

Rules:

- If it rains → ground becomes wet
- If ground is wet → shoes get dirty

Facts:

- It is raining.

Inference:

It rains → ground wet → shoes dirty

2. Backward Chaining (Goal-driven Reasoning)

Backward chaining starts with the goal and works backwards to check whether facts support the goal.

Working:

1. Identify the goal.
2. Find rules that produce that goal.
3. Check if their conditions are true
4. Continue until facts are verified.

Example:

Goal: shoes are dirty.

Check rule: ground wet → shoes dirty

Check: raining → ground wet

check fact: It is raining → TRUE → goal acheived.

Solution 2 : Hill climbing is a heuristic search algorithum used in AI for optimization problems. It continuously move towards a better state until no further improvement is possibl

It follows the principle :
"Always move towards the state with a higher value"

Working of Hill Climbing :
1. Start from an intial state.
2. Evaluate neighbouring states.
3. move to the best neighbour.
4. Repeat until no better neighbour exists.

Pseudo steps :
- Current = Initial State
- Loop =
    - Next = Best neighbor of current.
    - If Next is not better stop.
    - Else current = next.

Example : Finding max. value in mathematical function :-
The algorithum keeps climbing the graph until the highest peak is evaluated.

Applications :-
- Scheduling
- Game playing AI
- Robotics path planning
- Feature selection in ML.

Solution 3 : Symbolic reasoning is the process of manipulating symbols and logical rules to draw conclusions.
However, real-world problems contain certain or incomplete information.

Therefore AI uses reasoning under uncertainity.

Sources of uncertainity :-
- Incomplete knowledge
- Noisy data
- Ambigious language

Methods to solve uncertainity:

1. Probablity Theory :- uses likelihood of events
   Example: Probability (Rain tomorrow) = 70%.
   Used in : - prediction systems
   - ML
2. Bayesian Reasoning :- Updates belief after receiving evidence
   Formula : $P(H/E) = \frac{P(E/H)\,P(H)}{P(E)}$
3. Certainity Factor :- Respresents confidence level.
4. Fuzzy logic :- Handles vague statements

Solution 4:

1. Non-monoatomic Reasoning :-
   - once a conclusion is true → always true
   - Non-monootomic Reasoning allows AI to withdraw conclusions when new facts arrive.
   - Example :
     - Birds can fly.
     - Penguin is bird.
       → Penguin flies (wrong)

     Add new fact : Penguins cannot fly.

2. Statistical Reasoning :-
   - uses mathematical data and probability to make decisions. Instead of certainity → likelihood.

   Techniques:
   - Regression
   - Hypothesis Testing
   - Probability distributions
   - ML models

   - Ex: If 95% customers buy after ad → Predict new customer will buy.